

Notes: Stein (QJE, 1989)

Efficient Capital Markets, Inefficient Firms: A Model of Myopic Corporate Behavior

Contents

1 Big picture

2 Data and measurement (key objects)

- 2.1 This is a theory paper: the key objects are model primitives
- 2.2 Natural earnings
- 2.3 Persistent earnings component
- 2.4 Transitory earnings component
- 2.5 Observed earnings and hidden borrowing from the future
- 2.6 Cost function
- 2.7 What makes investment “invisible”
- 2.8 Stock price
- 2.9 Managerial horizon parameter
- 2.10 Managerial objective
- 2.11 Market-imputed natural earnings
- 2.12 Forecasting weights
- 2.13 Signal quality parameter

3 Models and empirical specifications (all equations + interpretation)

- 3.1 Equation (1): natural earnings are persistent plus transitory
- 3.2 Equation (2): reported earnings can be inflated by hidden borrowing
- 3.3 Equation (3): the efficient stock price
- 3.4 Equation (4): managerial preferences over current price and future earnings
- 3.5 Equation (5): rational filtering of earnings
- 3.6 Equation (6): future natural earnings are forecast from current and past reports

3.7	Equations (7) and (8): how informative is current earnings news?	
3.8	Why one extra dollar of current reported earnings raises price by a_0/r	
3.9	Equation (9): manager's first-order condition	
3.10	Equation (10): the signal-jamming equilibrium	
3.11	Why the market can be efficient and the firm still inefficient	
3.12	The no-myopia outcome is not a Nash equilibrium	
4	Identification and credibility (what makes the model persuasive)	
4.1	Why the paper is persuasive	
4.2	Assumption 1: invisible investment	
4.3	Assumption 2: managers must care about today's market price	
4.4	Assumption 3: earnings must predict future value	
4.5	Why positive stock-price reactions to investment announcements do not refute myopia	
4.6	Why start-up evidence does not kill the theory	
4.7	Why direct empirical tests are hard	
4.8	Capital market pressure: pro and con	
4.9	Integration and internal capital markets	
4.10	Analog Devices as an illustration	
4.11	Relation to signaling models	
4.12	Relation to noise-trader theories	
4.13	Main limitations	
5	Tables and figures	
5.1	Equation (2): the hidden-investment technology	
5.2	Equations (6) to (8): why current earnings matter for valuation	
5.3	Equation (10): the master result	
5.4	Section III.A: the myopia debate and capital expenditure announcements	
5.5	Section III.A again: why start-ups are a weak test case	
5.6	Section III.B: internal capital markets and the limits of integration	
6	Short summary	
7	Conclusion	
7.1	Core conclusion (what the paper establishes)	
7.2	Economic intuition	
7.3	Takeaway	

1 Big picture

- **Question.** Can managers become short-termist even when financial markets are fully rational and stock prices are efficient?
- **Conventional efficient-markets answer.** The textbook objection to “myopia” is simple: if investors understand that cutting good long-run investment lowers firm value, the stock market should punish such cuts rather than reward them. Therefore, a manager who wants a high stock price should not sacrifice good projects.
- **Stein’s answer.** That objection is incomplete. Even with a rational stock market, managers can still behave myopically if they can manipulate current reported earnings in ways outsiders cannot perfectly disentangle from genuine good news.
- **Core mechanism.** The market uses current earnings to forecast future firm value. Managers understand this, so they have an incentive to boost current earnings by “borrowing” from the future – equivalently, by cutting hidden long-run investment that depresses current accounting earnings.
- **Signal-jamming logic.** The market is not fooled on average. It correctly anticipates that managers will inflate current earnings and discounts reported earnings accordingly. But each manager takes that market conjecture as given and still wants to inflate current earnings at the margin. This is a signal-jamming equilibrium.
- **Why the result is striking.** The distortion does *not* require irrational investors. It arises in a rational-expectations equilibrium with efficient prices.
- **Managerial horizon parameter.** The paper summarizes short-horizon pressure with a parameter π , interpreted as the fraction of shares managers effectively have to sell at the current price. This can stand for takeover risk, compensation tied to current prices, outside shareholders’ liquidity needs, or the need to issue equity soon.
- **Main comparative static.** More pressure on current stock prices (higher π) makes managers more myopic.
- **Second key comparative static.** Myopia is worse when current earnings are more informative about future earnings. If the stock market heavily extrapolates from current earnings, managers get more price impact from inflating today’s bottom line.
- **Invisible investment is essential.** The paper is about investments that are hard for outsiders to observe separately from current operating costs – examples include customer relationships, pricing to build market share, advertising-like spending, sales effort, and some R&D or organizational investments.
- **Immediate implication.** Takeover pressure and low financial slack can have a real downside. They may force managers to cut good but hard-to-observe long-run investments in order to look good today.
- **What the paper does to the debate.** Stein directly challenges the claim that a positive stock-price reaction to investment announcements proves managers are not myopic. In his model, the opposite may be true: if managers use too high a hurdle rate, any project they do choose is especially good news.

- **Another prediction.** Start-up firms or R&D-intensive firms may face less of this specific myopia problem when current earnings are a weak signal of long-run value. That is why examples like Genentech do not refute the theory.
- **Novel organizational implication.** Internal capital markets can help because they create financial slack and reduce reliance on external equity. But integration also has limits: combining businesses can worsen aggregate earnings pressure if the combined firm’s current earnings become a stronger forecasting signal.
- **Takeaway.** Efficient stock markets do not automatically guarantee efficient firms. When managers care about current prices and can hide long-run investment inside current expenses, rational markets can coexist with systematically myopic corporate behavior.

2 Data and measurement (key objects)

2.1 This is a theory paper: the key objects are model primitives

There is no empirical sample, no regression dataset, and no identification design in the modern econometric sense. The paper’s “measurement” consists of the state variables, earnings process, pricing rule, and managerial objective.

2.2 Natural earnings

The firm’s unmanipulated or “natural” earnings are denoted

$$e_t^* = z_t + v_t, \tag{1}$$

where:

- z_t is a persistent component,
- v_t is a transitory noise component.

Interpretation. The market wants to learn the persistent part of earnings, because that is what matters for forecasting future value.

2.3 Persistent earnings component

The persistent component follows a random walk,

$$z_t = z_{t-1} + u_t, \tag{2}$$

with innovations u_t that are independent, mean zero normal shocks with variance σ_u^2 .

Interpretation. Permanent profitability evolves over time. A high current earnings realization may therefore contain news about future earnings.

2.4 Transitory earnings component

The transitory component v_t is independent over time, mean zero normal, with variance σ_v^2 .

Interpretation. This is temporary noise in earnings. When σ_v^2 is large, current earnings are a poor predictor of the future.

2.5 Observed earnings and hidden borrowing from the future

Managers can increase current reported earnings by choosing hidden borrowing b_t from next period's earnings. Observed earnings are

$$e_t = e_t^* + b_t - c(b_{t-1}), \quad (3)$$

where $c(\cdot)$ is a convex cost function.

Interpretation. “Borrowing” here is shorthand for sacrificing hidden long-run investment or liquidating assets too early. Current earnings rise today, but next period earnings fall.

2.6 Cost function

The function $c(\cdot)$ is convex and satisfies

$$c'(0) = 1 + r, \quad (4)$$

where r is the stock-market discount rate.

Interpretation. At the margin, moving one unit of earnings forward is value-neutral only at zero borrowing; convexity then makes larger borrowing inefficient. First-best firm value is maximized at $b_t = 0$.

2.7 What makes investment “invisible”

The paper is not about factories or other investments that accounting statements identify clearly. It is about expenditures that can be confused with ordinary operating costs.

Examples discussed by Stein include:

- customer coverage and relationship building,
- pricing aggressively today to build market share,
- some R&D or organizational spending,
- other outlays whose long-run payoff is real but whose current accounting treatment looks like an expense.

2.8 Stock price

Because the firm pays out earnings immediately as dividends, the ex-dividend stock price is the present value of expected future earnings:

$$P_t = E_t \sum_{j=1}^{\infty} \frac{e_{t+j}}{(1+r)^j}. \quad (5)$$

Interpretation. The stock price is fully rational conditional on the market's information. The market is efficient in the sense that it prices expected future cash flows correctly given what it sees.

2.9 Managerial horizon parameter

Managers care about today's stock price because they effectively liquidate a fraction π of their position each period. Stein interprets π flexibly as:

- takeover pressure,
- the need to issue new equity,
- compensation tied to current stock price,
- or other reasons current-market valuation matters.

2.10 Managerial objective

Managerial utility is

$$U_t = e_t + \pi P_t + (1 - \pi) \frac{e_{t+1}}{1+r}. \quad (6)$$

Interpretation. Managers care about current dividends, the current stock price on the fraction sold now, and next period's earnings on the shares they continue to hold. Because current choices affect only e_t and e_{t+1} in the model, later terms can be ignored.

2.11 Market-imputed natural earnings

In a steady state where the market expects a constant amount of borrowing b every period, it backs out imputed natural earnings as

$$e_t^n = e_t + c(b) - b. \quad (7)$$

Interpretation. The market rationally removes the expected average manipulation from observed earnings.

2.12 Forecasting weights

Future natural earnings are forecast using a distributed lag,

$$E_{t+k}^{bigl}(e_{t+k}^*) = \sum_{j=0}^{\infty} a_j e_{t-j}^n, \quad k > 0, \quad (8)$$

where the weights a_j sum to one.

Interpretation. The coefficient a_0 is especially important because it tells how much the market updates its forecast of the future when current earnings go up by one dollar.

2.13 Signal quality parameter

Stein uses the random-walk-plus-noise filter to obtain

$$a_0 = \left(\frac{k^2}{4} + k \right)^{1/2} - \frac{k}{2}, \quad (9)$$

where

$$k = \frac{\sigma_u^2}{\sigma_v^2}. \quad (10)$$

Interpretation.

- If permanent shocks are small relative to transitory noise, then k is small and a_0 is near zero.
- If current earnings reveal a lot about the persistent component, then k is large and a_0 is near one.

This is the paper's formal measure of how informative current earnings are about the future.

3 Models and empirical specifications (all equations + interpretation)

3.1 Equation (1): natural earnings are persistent plus transitory

$$e_t^* = z_t + v_t. \quad (1)$$

Interpretation.

- The market cannot directly see whether a good earnings number reflects a durable improvement (z_t) or a temporary fluctuation (v_t).
- This inference problem is what gives managers room to manipulate.

3.2 Equation (2): reported earnings can be inflated by hidden borrowing

$$e_t = e_t^* + b_t - c(b_{t-1}). \quad (2)$$

Interpretation.

- b_t raises current reported earnings today.
- The cost $c(b_{t-1})$ hits next period's earnings.

- The function is convex, so pushing too much income forward becomes increasingly expensive.

The key economic step is that outsiders do not directly observe b_t . If they could, the market could perfectly separate real good news from earnings inflation.

3.3 Equation (3): the efficient stock price

$$P_t = E_t \sum_{j=1}^{\infty} \frac{e_{t+j}}{(1+r)^j}. \quad (3)$$

Interpretation.

- This is a fully rational pricing equation.
- The market does not have to be naive for the myopia result to go through.
- The whole point of the paper is that efficient prices can coexist with inefficient corporate decisions.

3.4 Equation (4): managerial preferences over current price and future earnings

$$U_t = e_t + \pi P_t + (1 - \pi) \frac{e_{t+1}}{1+r}. \quad (4)$$

Interpretation.

- A higher π means a shorter effective horizon.
- If $\pi = 0$, managers care only about the present value of earnings and there is no myopia.
- If π is high, current stock-price effects loom large relative to next period's earnings loss.

3.5 Equation (5): rational filtering of earnings

In steady state, the market expects constant borrowing b and therefore reconstructs natural earnings as

$$e_t^n = e_t + c(b) - b. \quad (5)$$

Interpretation. The market is not systematically fooled. It strips out the average amount of expected manipulation when processing earnings news.

3.6 Equation (6): future natural earnings are forecast from current and past reports

$$E_t(e_{t+k}^*) = \sum_{j=0}^{\infty} a_j e_{t-j}^n, \quad k > 0. \quad (6)$$

Interpretation.

- The signal-jamming incentive comes from a_0 , the weight on current earnings.
- When current earnings move the forecast of all future earnings, current price becomes sensitive to hidden manipulation.

3.7 Equations (7) and (8): how informative is current earnings news?

$$a_0 = \left(\frac{k^2}{4} + k \right)^{1/2} - \frac{k}{2}, \quad (7)$$

$$k = \frac{\sigma_u^2}{\sigma_v^2}. \quad (8)$$

Interpretation.

- High a_0 means the market reads current earnings as persistent news.
- Low a_0 means the market treats current earnings largely as noise.
- This is why myopia should be strongest in mature businesses where current profits are informative and weakest when current profits say little about long-run success.

3.8 Why one extra dollar of current reported earnings raises price by a_0/r

Holding the market's conjecture of b fixed, an extra dollar of current reported earnings raises imputed natural earnings one-for-one. The market then revises expected future earnings upward by a_0 each period. Capitalizing that constant revision gives

$$\frac{dP_t}{db_t} = \frac{a_0}{r}. \quad (11)$$

Interpretation. This is the fundamental price benefit from dressing up current earnings.

3.9 Equation (9): manager's first-order condition

The manager chooses b_t taking the market's conjecture as fixed:

$$\frac{dU_t}{db_t} = \frac{de_t}{db_t} + \pi \frac{dP_t}{db_t} + \frac{1 - \pi}{1 + r} \frac{de_{t+1}}{db_t} = 0. \quad (9)$$

Using

$$\frac{de_t}{db_t} = 1, \quad \frac{de_{t+1}}{db_t} = -c'(b_t), \quad \frac{dP_t}{db_t} = \frac{a_0}{r}, \quad (12)$$

we get the steady-state equilibrium condition.

3.10 Equation (10): the signal-jamming equilibrium

$$c'(b) = \frac{1+r}{1-\pi} \left(1 + \pi \frac{a_0}{r} \right). \quad (10)$$

Interpretation.

- If $\pi = 0$, then $c'(b) = 1 + r$, so with $c'(0) = 1 + r$ the optimum is $b = 0$. No myopia.
- If $\pi > 0$, then the right-hand side exceeds $1 + r$, so $b > 0$. Managers inflate current earnings by borrowing from the future.
- A larger π increases b : more short-horizon pressure means more myopia.
- A larger a_0 increases b : the more the market extrapolates from current earnings, the stronger the incentive to manipulate them.

3.11 Why the market can be efficient and the firm still inefficient

Equation (10) is the paper's punchline. The market correctly anticipates the equilibrium b and prices the firm rationally. Yet managers still choose too much b because, at the margin, they compare the private benefit from a higher current price with only the fraction $(1 - \pi)$ of next period's cost that they personally internalize.

Economic intuition. The market is right in equilibrium, but each manager is still trapped into playing a bad strategy against the market's fixed beliefs. This is the prisoner's-dilemma flavor of the model.

3.12 The no-myopia outcome is not a Nash equilibrium

Suppose the market expected no manipulation at all. Then current earnings would be treated as especially informative, which would make it very attractive for managers to inflate them. So the cooperative outcome of no jamming cannot be sustained.

Interpretation. "Managers cannot systematically fool the market" does *not* imply "managers will not try." They try in equilibrium, and the attempt itself destroys value.

4 Identification and credibility (what makes the model persuasive)

4.1 Why the paper is persuasive

The model is persuasive precisely because it grants a great deal to the efficient-markets view. Investors are rational. Prices are efficient conditional on information. There is no need for behavioral mistakes. If myopia still appears, the mechanism is conceptually important.

4.2 Assumption 1: invisible investment

The result depends on some managerial actions being hard for outsiders to observe directly.

- If the market could separately observe long-run investment and current waste, it would not reward earnings inflation.
- The more hidden the investment margin, the stronger the signal-jamming problem.

Interpretation. This is not a theory of all corporate investment. It is a theory of the subset of investments that accounting statements summarize poorly.

4.3 Assumption 2: managers must care about today's market price

The parameter π is the reduced-form source of short-horizon pressure. Stein offers several interpretations:

- takeover risk,
- the need to issue equity,
- stock-based compensation,
- or shareholders' liquidity needs.

Interpretation. The theory is not tied to one institutional story. Any force that makes today's stock price matter can create the distortion.

4.4 Assumption 3: earnings must predict future value

The stock-price benefit from manipulation depends on a_0 .

- If current earnings say almost nothing about future profitability, then inflating them does little for the price.
- If current earnings are highly informative, then price pressure becomes severe.

Interpretation. This delivers a clean cross-sectional prediction: mature firms with persistent earnings should face more of this problem than early-stage firms whose value depends on distant uncertain breakthroughs.

4.5 Why positive stock-price reactions to investment announcements do not refute myopia

A central contribution of the paper is to overturn a common argument in the debate. Suppose myopic managers set too high a hurdle rate for investment. Then any project that *does* get announced must be especially good.

Implication.

- The market can rationally react positively to announced investment.

- Yet managers may still be underinvesting overall because many merely good projects never get chosen.

Economic lesson. Announcement returns identify the quality of accepted projects, not whether the firm is leaving many positive-NPV projects on the table.

4.6 Why start-up evidence does not kill the theory

The fact that some start-ups can earn little or nothing today and still command high valuations is not inconsistent with the model.

- For such firms, current earnings may carry very little information about future value.
- That means a_0 is low, so there is less gain from inflating current earnings.

Interpretation. The paper predicts heterogeneity in myopia across firms, not universal short-termism.

4.7 Why direct empirical tests are hard

Stein explicitly notes that the most important forms of myopic behavior may be difficult to detect in conventional accounting data.

- Expenditures on plant and equipment are often visible and therefore less likely to be sacrificed.
- Hidden investments inside operating expenses are harder to identify and therefore more vulnerable.

Interpretation. This is a subtle but important point: the absence of dramatic capital-expenditure cuts is not decisive evidence against myopia.

4.8 Capital market pressure: pro and con

The model gives a more balanced view of takeover threats and low financial slack than a pure agency story.

- **Agency view:** capital market pressure disciplines managers and cuts waste.
- **Stein's view:** the same pressure can also cut good long-run investment when outsiders cannot see it clearly.

Interpretation. Capital market pressure is not automatically good or bad. Its welfare effect depends on how much managerial discretion hides negative-NPV pet projects versus positive-NPV invisible investment.

4.9 Integration and internal capital markets

The paper gives a nuanced view of conglomeration and internal capital markets.

- Internal cash flow creates financial slack and lowers dependence on outside equity. That reduces effective π and can mitigate myopia.

- But integration can also increase the pressure to deliver smooth consolidated earnings if the combined firm's current earnings become a stronger signal about future value.

Interpretation. There can be both a benefit and a limit to internal capital markets. A cash-generating mature unit may help fund a cash-using innovative unit, but it may also impose a stronger short-run earnings culture on that innovative activity.

4.10 Analog Devices as an illustration

Stein discusses Analog Devices' venture-capital-style structure as suggestive evidence that firms sometimes separate innovative projects from the main operating company for reporting reasons.

Interpretation. The organizational form of innovation financing can be shaped not only by technological or contractual issues, but also by how accounting treatment feeds into stock-market scrutiny.

4.11 Relation to signaling models

Stein contrasts the present paper with his earlier signaling-based work and with Miller-Rock style dividend signaling.

- **Signal-jamming here:** no ex ante private type information is needed; the hidden action itself creates the problem.
- **Signaling models:** firms distort actions to separate good types from bad types.

Interpretation. The two approaches are complementary. Myopia can arise either because actions are hidden or because types are hidden.

4.12 Relation to noise-trader theories

The paper also differs from later or parallel theories in which inefficient stock prices create short-termism.

Interpretation. Stein's mechanism is stronger in one sense: even if stock prices are fully rational, short-term price pressure can still distort real decisions.

4.13 Main limitations

The model is intentionally stylized.

- It is a steady-state linear-normal learning model.
- It compresses many institutional details into the reduced-form horizon parameter π .
- It models only one-period borrowing from future earnings.
- It does not offer a direct empirical implementation or measurement strategy for hidden investment.

Interpretation. These are real simplifications, but they are also what make the central mechanism transparent.

5 Tables and figures

This is a **theory paper** with no regression tables and no central diagrams. The key things to “read” are the model’s equations and the discussion examples in Section III.

5.1 Equation (2): the hidden-investment technology

Read:

- Reported earnings equal natural earnings plus current borrowing minus the repayment cost from last period’s borrowing.
- The market sees e_t but not b_t .
- Convexity of $c(\cdot)$ makes aggressive earnings inflation increasingly costly.

Economic message. Myopia is not just “managers liking short-term numbers.” The model gives a concrete technology through which numbers can be dressed up at the expense of real future value.

5.2 Equations (6) to (8): why current earnings matter for valuation

Read:

- The market filters current and past earnings to infer future natural earnings.
- a_0 is the marginal weight on current earnings.
- a_0 rises when permanent earnings variation is large relative to transitory noise.

Economic message. The stock market is rational, but it still has to solve an inference problem. The informativeness of current earnings determines the strength of short-run price pressure.

5.3 Equation (10): the master result

Read:

- The left-hand side is the marginal real cost of borrowing from future earnings.
- The right-hand side rises with π and with a_0 .
- Therefore equilibrium borrowing b rises with both short-horizon pressure and earnings informativeness.

Economic message. This one equation contains the whole paper. It shows exactly when efficient markets generate inefficient managerial behavior.

5.4 Section III.A: the myopia debate and capital expenditure announcements

Read:

- Stein argues that positive market reactions to investment announcements do not reject myopia.
- In a myopic equilibrium, managers use a hurdle rate that is too high.
- So any accepted project is unusually strong news.

Economic message. Event-study evidence on *accepted* projects is not enough to measure the unseen set of *rejected* positive-NPV projects.

5.5 Section III.A again: why start-ups are a weak test case

Read:

- Young high-tech firms can have low current earnings and still high valuations.
- Stein's explanation is that current earnings for such firms are poor predictors of the future.
- That means a_0 is low, so there is little incentive to manipulate today's earnings.

Economic message. Cross-firm variation in information structure matters. Mature firms and start-ups should not be expected to exhibit the same kind of short-termism.

5.6 Section III.B: internal capital markets and the limits of integration

Read:

- Financial slack lowers the need to impress outside equity markets.
- But integration can worsen signal-jamming if consolidated earnings become a more important valuation signal.
- This can bias resource allocation away from units whose value depends on hard-to-measure long-run investment.

Economic message. Internal capital markets are not unambiguously good. They can soften external financing pressure while simultaneously importing short-run earnings pressure across divisions.

6 Short summary

Stein develops a rational-expectations model in which managers inflate current earnings by sacrificing hidden long-run investment, even though the stock market is efficient and correctly anticipates this behavior. The distortion arises because managers care about today's stock price and take the market's conjectures as fixed. In equilibrium, rational investors discount expected earnings inflation, but managers still choose too much of it at the margin. Myopia is stronger when managers face more short-horizon pressure and when current earnings are more informative about future earnings. The framework helps reinterpret evidence from takeover debates, investment announcement studies, and the organization of internal capital markets.

7 Conclusion

7.1 Core conclusion (what the paper establishes)

The paper establishes that efficient stock prices do not by themselves guarantee efficient corporate behavior. If managers care about current stock prices and can manipulate reported earnings by cutting hidden long-run investment, a rational stock market can coexist with systematically myopic firms.

7.2 Economic intuition

Managers want the market to extrapolate from current earnings, so they try to make current earnings look strong. The market knows this and discounts expected manipulation, but each manager still has an incentive to manipulate at the margin because current price gains are partly privatized while future earnings losses are only partly internalized. The equilibrium is therefore rational but inefficient.

7.3 Takeaway

Stein's central insight is compact and durable: *"the market is not fooled" is not the same as "managers have no incentive to try to fool it."* Once some investments are hard to observe and current stock prices matter enough, short-term performance pressure can distort real decisions even in a world of rational investors and efficient capital markets.